

LAB-02

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Task 1

```
student@6312vm-csr02:~$ docker run -it alpine
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
6a0ac1617861: Pull complete
Digest: sha256:5b10f432ef3dalb8d4c7eb6c487f2f5a8f096bc91145e68878dd4a5019afde11
Status: Downloaded newer image for alpine:latest
/ # apk update
v3.23.4-139-gfc5e77717a8 [https://dl-cdn.alpinelinux.org/alpine/v3.23/main]
v3.23.4-138-gc45b849a722 [https://dl-cdn.alpinelinux.org/alpine/v3.23/community]
OK: 27578 distinct packages available
/ # apk add vim
(1/5) Installing vim-common (9.2.0437-r0)
(2/5) Installing xxd (9.2.0437-r0)
(3/5) Installing ncurses-terminfo-base (6.5_p20251123-r0)
(4/5) Installing libncursesw (6.5_p20251123-r0)
(5/5) Installing vim (9.2.0437-r0)
Executing busybox-1.37.0-r30.trigger
OK: 38.0 MiB in 21 packages
/ # mkdir app
/ # cd app
/app # vim hello.py
/app # cat hello.py
print("Hello World From Alpine Container")
/app # python3 hello.py
/bin/sh: python3: not found
/app # apk add python3
( 1/15) Installing libbz2 (1.0.8-r6)
( 2/15) Installing libexpat (2.7.5-r0)
( 3/15) Installing libffi (3.5.2-r0)
( 4/15) Installing gdbm (1.26-r0)
( 5/15) Installing xz-libs (5.8.3-r0)
( 6/15) Installing libgcc (15.2.0-r2)
( 7/15) Installing libstdc++ (15.2.0-r2)
( 8/15) Installing mpdecimal (4.0.1-r0)
( 9/15) Installing libpanelw (6.5_p20251123-r0)
(10/15) Installing readline (8.3.1-r0)
(11/15) Installing sqlite-libs (3.51.2-r0)
(12/15) Installing python3 (3.12.13-r0)
(13/15) Installing python3-pycache-pyc0 (3.12.13-r0)
(14/15) Installing pyc (3.12.13-r0)
(15/15) Installing python3-pyc (3.12.13-r0)
Executing busybox-1.37.0-r30.trigger
OK: 76.6 MiB in 36 packages
/app # python3 hello.py
Hello World From Alpine Container
/app # █
```

Task 2

```
python/docker=dpo Docker image for local development of http://...
student@6312vm-csr02:~$ docker run -it python:3.6
Unable to find image 'python:3.6' locally
3.6: Pulling from library/python
0e29546d541c: Pull complete
9b829c73b52b: Pull complete
cb5b7ae36172: Pull complete
6494e4811622: Pull complete
6f9f74896dfa: Pull complete
5e3b1213efc5: Pull complete
9fddfdc56334: Pull complete
404f02044bac: Pull complete
c4f42be2be53: Pull complete
Digest: sha256:f8652afaf88c25f0d22354d547d892591067aa4026a7fa9a6819df9f300af6fc
Status: Downloaded newer image for python:3.6
Python 3.6.15 (default, Dec 21 2021, 12:03:22)
[GCC 10.2.1 20210110] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Testing Python 3.6")
Testing Python 3.6
>>> exit
Use exit() or Ctrl-D (i.e. EOF) to exit
>>> exit()
student@6312vm-csr02:~$ docker run -it python:3.6 /bin/bash
root@3d1a36e8cbd4:/# apt-get update
Get:1 http://security.debian.org/debian-security bullseye-security InRelease [27.2 kB]
Get:2 http://deb.debian.org/debian bullseye InRelease [75.1 kB]
Get:3 http://deb.debian.org/debian bullseye-updates InRelease [44.0 kB]
Get:4 http://deb.debian.org/debian bullseye/main amd64 Packages [8066 kB]
Get:5 http://deb.debian.org/debian bullseye-updates/main amd64 Packages [18.8 kB]
Get:6 http://security.debian.org/debian-security bullseye-security/main amd64 Packages [451 kB]
Fetched 8682 kB in 3s (2610 kB/s)
Reading package lists... Done
root@3d1a36e8cbd4:/# apt-get install -y vim
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libgpm2 vim-common vim-runtime xxd
Suggested packages:
  gpm ctags vim-doc vim-scripts
The following NEW packages will be installed:
  libgpm2 vim vim-common vim-runtime xxd
0 upgraded, 5 newly installed, 0 to remove and 202 not upgraded.
Need to get 8179 kB of archives.
After this operation, 36.9 MB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bullseye/main amd64 libgpm2 amd64 1.20.7-8 [35.6 kB]
Get:2 http://security.debian.org/debian-security bullseye-security/main amd64 xxd amd64 2:8.2.2434-3+deb11u3 [193
]
Get:3 http://security.debian.org/debian-security bullseye-security/main amd64 vim-common all 2:8.2.2434-3+deb11u3
27 kB]
Get:4 http://security.debian.org/debian-security bullseye-security/main amd64 vim-runtime all 2:8.2.2434-3+deb11u
6229 kB]
Get:5 http://security.debian.org/debian-security bullseye-security/main amd64 vim amd64 2:8.2.2434-3+deb11u3 [149
B]
Fetched 8179 kB in 1s (9720 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package xxd.
```

```

Suggested packages:
  gpm ctags vim-doc vim-scripts
The following NEW packages will be installed:
  libgpm2 vim vim-common vim-runtime xxd
0 upgraded, 5 newly installed, 0 to remove and 202 not upgraded.
Need to get 8179 kB of archives.
After this operation, 36.9 MB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bullseye/main amd64 libgpm2 amd64 1.20.7-8 [35.6 kB]
Get:2 http://security.debian.org/debian-security bullseye-security/main amd64 xxd amd64 2:8.2.2434-3+deb11u3 [193 kB]
Get:3 http://security.debian.org/debian-security bullseye-security/main amd64 vim-common all 2:8.2.2434-3+deb11u3 [27 kB]
Get:4 http://security.debian.org/debian-security bullseye-security/main amd64 vim-runtime all 2:8.2.2434-3+deb11u3 [6229 kB]
Get:5 http://security.debian.org/debian-security bullseye-security/main amd64 vim amd64 2:8.2.2434-3+deb11u3 [1496 kB]
Fetched 8179 kB in 1s (9720 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package xxd.
(Reading database ... 23373 files and directories currently installed.)
Preparing to unpack .../xxd_2%3a8.2.2434-3+deb11u3_amd64.deb ...
Unpacking xxd (2:8.2.2434-3+deb11u3) ...
Selecting previously unselected package vim-common.
Preparing to unpack .../vim-common_2%3a8.2.2434-3+deb11u3_all.deb ...
Unpacking vim-common (2:8.2.2434-3+deb11u3) ...
Selecting previously unselected package libgpm2:amd64.
Preparing to unpack .../libgpm2_1.20.7-8_amd64.deb ...
Unpacking libgpm2:amd64 (1.20.7-8) ...
Selecting previously unselected package vim-runtime.
Preparing to unpack .../vim-runtime_2%3a8.2.2434-3+deb11u3_all.deb ...
Adding 'diversion of /usr/share/vim/vim82/doc/help.txt to /usr/share/vim/vim82/doc/help.txt.vim-tiny by vim-runtime'
Adding 'diversion of /usr/share/vim/vim82/doc/tags to /usr/share/vim/vim82/doc/tags.vim-tiny by vim-runtime'
Unpacking vim-runtime (2:8.2.2434-3+deb11u3) ...
Selecting previously unselected package vim.
Preparing to unpack .../vim_2%3a8.2.2434-3+deb11u3_amd64.deb ...
Unpacking vim (2:8.2.2434-3+deb11u3) ...
Setting up libgpm2:amd64 (1.20.7-8) ...
Setting up xxd (2:8.2.2434-3+deb11u3) ...
Setting up vim-common (2:8.2.2434-3+deb11u3) ...
Setting up vim-runtime (2:8.2.2434-3+deb11u3) ...
Setting up vim (2:8.2.2434-3+deb11u3) ...
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/vim (vim) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/vimdiff (vimdiff) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/rvim (rvim) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/rview (rview) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/vi (vi) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/view (view) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/ex (ex) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/editor (editor) in auto mode
Processing triggers for libc-bin (2.31-13+deb11u2) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
root@3dla36e8cbd4:/# mkdir app
root@3dla36e8cbd4:/# cd app
root@3dla36e8cbd4:/app# vim hello.py
root@3dla36e8cbd4:/app# python3 hello.py
Hello World from Python3 Container
root@3dla36e8cbd4:/app# █

```

Task 3

```
student@6312vm-csr02:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
540a22252dl9   linux-lab_ubuntu "/bin/bash"             4 days ago    Up 4 days                    my-ubuntu
student@6312vm-csr02:~$ git clone --single-branch --branch dist-amd64 https://github.com/tianon/docker-brew-ubuntu-core.git
Cloning into 'docker-brew-ubuntu-core'...
remote: Enumerating objects: 507, done.
remote: Counting objects: 100% (30/30), done.
remote: Compressing objects: 100% (26/26), done.
remote: Total 507 (delta 4), reused 14 (delta 3), pack-reused 477 (from 1)
Receiving objects: 100% (507/507), 212.74 MiB | 10.47 MiB/s, done.
Resolving deltas: 100% (186/186), done.
student@6312vm-csr02:~$ cd docker-brew-ubuntu-core/xenial
student@6312vm-csr02:~/docker-brew-ubuntu-core/xenial$ ls .
alias      build-info.txt  MD5SUMS        SHA1SUMS        SHA256SUMS      ubuntu-xenial-core-cloudimg-amd64.manifest
arches     Dockerfile     MD5SUMS.gpg    SHA1SUMS.gpg    SHA256SUMS.gpg  ubuntu-xenial-core-cloudimg-amd64-root.tar.gz
student@6312vm-csr02:~/docker-brew-ubuntu-core/xenial$ docker image build .
Sending build context to Docker daemon 46.01MB
Step 1/6 : FROM scratch
-->
Step 2/6 : ADD ubuntu-xenial-core-cloudimg-amd64-root.tar.gz /
--> 27e52787f861
Step 3/6 : RUN set -xe          && echo '#!/bin/sh' > /usr/sbin/policy-rc.d          && echo 'exit 101' >> /usr/sbin/policy-rc.d
&& chmod +x /usr/sbin/policy-rc.d          && dpkg-divert --local --rename --add /sbin/initctl
&& cp -a /usr/sbin/policy-rc.d /sbin/initctl          && sed -i 's/'exit.*/exit 0/' /sbin/initctl          && echo 'for
ce-unsafe-io' > /etc/dpkg/dpkg.cfg.d/docker-apt-speedup          && echo 'DPkg::Post-Invoke { "rm -f /var/cache
/apt/archives/*.deb /var/cache/apt/archives/partial/*.deb /var/cache/apt/*.bin || true"; };' > /etc/apt/apt.conf.d
/docker-clean          && echo 'APT::Update::Post-Invoke { "rm -f /var/cache/apt/archives/*.deb /var/cache/apt/archives/par
tial/*.deb /var/cache/apt/*.bin || true"; };' >> /etc/apt/apt.conf.d/docker-clean          && echo 'Dir::Cache::pkgcach
e ""; Dir::Cache::srcpkgcache "";' >> /etc/apt/apt.conf.d/docker-clean          && echo 'Acquire::Languages "none";'
> /etc/apt/apt.conf.d/docker-no-languages          && echo 'Acquire::GzipIndexes "true"; Acquire::CompressionTy
pes::Order:: "gz";' > /etc/apt/apt.conf.d/docker-gzip-indexes          && echo 'Apt::AutoRemove::SuggestsImportant
"false";' > /etc/apt/apt.conf.d/docker-autoremove-suggests
--> Running in 672b3c6b8c1d
+ echo #!/bin/sh
+ echo exit 101
+ chmod +x /usr/sbin/policy-rc.d
+ dpkg-divert --local --rename --add /sbin/initctl
Adding 'local diversion of /sbin/initctl to /sbin/initctl.distrib'
+ cp -a /usr/sbin/policy-rc.d /sbin/initctl
+ sed -i 's/'exit.*/exit 0/' /sbin/initctl
+ echo force-unsafe-io
+ echo DPkg::Post-Invoke { "rm -f /var/cache/apt/archives/*.deb /var/cache/apt/archives/partial/*.deb /var/cache/apt
/*.bin || true"; };
+ echo APT::Update::Post-Invoke { "rm -f /var/cache/apt/archives/*.deb /var/cache/apt/archives/partial/*.deb /var/ca
che/apt/*.bin || true"; };
+ echo Dir::Cache::pkgcache ""; Dir::Cache::srcpkgcache "";
+ echo Acquire::Languages "none";
+ echo Acquire::GzipIndexes "true"; Acquire::CompressionTypes::Order:: "gz";
+ echo Apt::AutoRemove::SuggestsImportant "false";
Removing intermediate container 672b3c6b8c1d
--> 62beea2c4c5c
Step 4/6 : RUN rm -rf /var/lib/apt/lists/*
--> Running in 7bfa7623bae6
Removing intermediate container 7bfa7623bae6
--> a392ca75d43f
Step 5/6 : RUN mkdir -p /run/systemd && echo 'docker' > /run/systemd/container
```

```
Step 5/6 : RUN mkdir -p /run/systemd && echo 'docker' > /run/systemd/container
--> Running in f3aadc6de546
Removing intermediate container f3aadc6de546
--> 31e2fe482848
Step 6/6 : CMD ["/bin/bash"]
--> Running in c59aalc9e51d
Removing intermediate container c59aalc9e51d
--> 2b54951e0061
Successfully built 2b54951e0061
student@6312vm-csr02:~/docker-brew-ubuntu-core/xenial$ docker images
REPOSITORY      TAG          IMAGE ID       CREATED        SIZE
<none>          <none>       2b54951e0061   24 seconds ago 135MB
linux-lab_ubuntu latest      ed1bbc491458   4 days ago    335MB
ubuntu          18.04       f9a80a55f492   2 years ago    63.2MB
student@6312vm-csr02:~/docker-brew-ubuntu-core/xenial$ docker run -ti 2b54951e0061
root@4eb6fba9c250:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys usr var
root@4eb6fba9c250:/# exit
exit
student@6312vm-csr02:~/docker-brew-ubuntu-core/xenial$ cd ..
student@6312vm-csr02:~/docker-brew-ubuntu-core$
```

Task 4

```
student@6312vm-csr02:~$ ls
Desktop  docker-brew-ubuntu-core  Documents  Downloads  Music  Pictures  Public  Templates  Videos
student@6312vm-csr02:~$ mkdir custom-container && cd custom-container
student@6312vm-csr02:~/custom-container$ vim helloinwk.py
student@6312vm-csr02:~/custom-container$ vim Dockerfile
student@6312vm-csr02:~/custom-container$ ls
Dockerfile  helloinwk.py
student@6312vm-csr02:~/custom-container$ docker image build .
Sending build context to Docker daemon  3.072kB
Step 1/6 : FROM ubuntu
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335ddd46: Pull complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
--> 30ba44506a6d
Step 2/6 : RUN apt-get update
--> Running in 3701e7b52957
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [26.2 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [34.4 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1874 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [20.1 MB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [69.5 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [34.4 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [34.7 kB]
Fetched 23.3 MB in 3s (6791 kB/s)
Reading package lists...
Removing intermediate container 3701e7b52957
--> 4ef568941f80
Step 3/6 : RUN apt-get -y install python3
--> Running in 83b24f039a55
Reading package lists...
Building dependency tree...
Reading state information...
Solving dependencies...
The following additional packages will be installed:
  ca-certificates libexpat1 libffi8 libpython3-stdlib libpython3.14-minimal
  libpython3.14-stdlib libreadline8t64 libsqlite3-0 media-types netbase
  openssl python3-minimal python3.14 python3.14-minimal readline-common tzdata
Suggested packages:
  python3-doc python3-tk python3-venv python3.14-venv python3.14-doc binutils
  binfmt-support readline-doc
The following NEW packages will be installed:
  ca-certificates libexpat1 libffi8 libpython3-stdlib libpython3.14-minimal
  libpython3.14-stdlib libreadline8t64 libsqlite3-0 media-types netbase
  openssl python3 python3-minimal python3.14 python3.14-minimal
  readline-common tzdata
0 upgraded, 17 newly installed, 0 to remove and 1 not upgraded.
Need to get 9495 kB of archives.
After this operation, 32.3 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libexpat1 amd64 2.7.4-1 [94.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libpython3.14-minimal amd64 3.14.4-1 [916 kB]
```

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Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libexpat1 amd64 2.7.4-1 [94.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libpython3.14-minimal amd64 3.14.4-1 [916 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 python3.14-minimal amd64 3.14.4-1 [2579 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 python3-minimal amd64 3.14.3-0ubuntu2 [25.8 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 media-types all 14.0.0build1 [31.4 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 netbase all 6.5build1 [13.0 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 tzdata all 2026a-3ubuntu1 [194 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 libffi8 amd64 3.5.2-4 [26.3 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 readline-common all 8.3-4 [61.5 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 libreadline8t64 amd64 8.3-4 [164 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsqlite3-0 amd64 3.46.1-9 [722 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/main amd64 libpython3.14-stdlib amd64 3.14.4-1 [2406 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute/main amd64 python3.14 amd64 3.14.4-1 [830 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute/main amd64 libpython3-stdlib amd64 3.14.3-0ubuntu2 [8476 B]
Get:15 http://archive.ubuntu.com/ubuntu resolute/main amd64 python3 amd64 3.14.3-0ubuntu2 [22.9 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute/main amd64 openssl amd64 3.5.5-1ubuntu3 [1243 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute/main amd64 ca-certificates all 20260223 [158 kB]
debconf: unable to initialize frontend: Dialog
debconf: (TERM is not set, so the dialog frontend is not usable.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-gnu/perl5/5.40 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8, <STDIN> line 17.)
debconf: falling back to frontend: Teletype
debconf: unable to initialize frontend: Teletype
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Noninteractive
Preconfiguring packages ...
Fetched 9495 kB in 1s (10.9 MB/s)
Selecting previously unselected package libexpat1:amd64.
(Reading database ... 7724 files and directories currently installed.)
Preparing to unpack .../libexpat1_2.7.4-1_amd64.deb ...
Unpacking libexpat1:amd64 (2.7.4-1) ...
Selecting previously unselected package libpython3.14-minimal:amd64.
Preparing to unpack .../libpython3.14-minimal_3.14.4-1_amd64.deb ...
Unpacking libpython3.14-minimal:amd64 (3.14.4-1) ...
Selecting previously unselected package python3.14-minimal.
Preparing to unpack .../python3.14-minimal_3.14.4-1_amd64.deb ...
Unpacking python3.14-minimal (3.14.4-1) ...
Setting up libpython3.14-minimal:amd64 (3.14.4-1) ...
Setting up libexpat1:amd64 (2.7.4-1) ...
Setting up python3.14-minimal (3.14.4-1) ...
Selecting previously unselected package python3-minimal.
(Reading database ... 8078 files and directories currently installed.)
Preparing to unpack .../00-python3-minimal_3.14.3-0ubuntu2_amd64.deb ...
Unpacking python3-minimal (3.14.3-0ubuntu2) ...
Selecting previously unselected package media-types.
Preparing to unpack .../01-media-types_14.0.0build1_all.deb ...
Unpacking media-types (14.0.0build1) ...
Selecting previously unselected package netbase.
Preparing to unpack .../02-netbase_6.5build1_all.deb ...
Unpacking netbase (6.5build1) ...
Selecting previously unselected package tzdata.
Preparing to unpack .../03-tzdata_2026a-3ubuntu1_all.deb ...
Unpacking tzdata (2026a-3ubuntu1) ...
Selecting previously unselected package libffi8:amd64.

```



```

debconf: unable to initialize frontend: Teletype
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Noninteractive

Current default time zone: 'Etc/UTC'
Local time is now:      Wed May  6 17:07:31 UTC 2026.
Universal Time is now:  Wed May  6 17:07:31 UTC 2026.
Run 'dpkg-reconfigure tzdata' if you wish to change it.

Setting up libffi8:amd64 (3.5.2-4) ...
Setting up netbase (6.5build1) ...
Setting up openssl (3.5.5-1ubuntu3) ...
Setting up readline-common (8.3-4) ...
Setting up ca-certificates (20260223) ...
debconf: unable to initialize frontend: Dialog
debconf: (TERM is not set, so the dialog frontend is not usable.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-gnu/perl5/5.40 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
debconf: unable to initialize frontend: Teletype
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Noninteractive
Updating certificates in /etc/ssl/certs...
144 added, 0 removed; done.
Setting up libreadline8t64:amd64 (8.3-4) ...
Setting up libpython3.14-stdlib:amd64 (3.14.4-1) ...
Setting up python3.14 (3.14.4-1) ...
Setting up libpython3-stdlib:amd64 (3.14.3-0ubuntu2) ...
Setting up python3 (3.14.3-0ubuntu2) ...
running python rtupdate hooks for python3.14...
running python post-rtupdate hooks for python3.14...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
Processing triggers for ca-certificates (20260223) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Removing intermediate container 83b24f039a55
--> 01bd5a17f305
Step 4/6 : COPY helloinwk.py /helloinwk.py
--> 2000229d2ab1
Step 5/6 : RUN ["chmod", "+x", "/helloinwk.py"]
--> Running in 448a6a9bf0af
Removing intermediate container 448a6a9bf0af
--> 3f8c6a78f64e
Step 6/6 : CMD ["/helloinwk.py"]
--> Running in a3afbe0978fd
Removing intermediate container a3afbe0978fd
--> 31d61f0fde13
Successfully built 31d61f0fde13
student@6312vm-csr02:~/custom-container$ docker container run 31d61f0fde13
Hello from Internetworking Class!
student@6312vm-csr02:~/custom-container$ cd ..
student@6312vm-csr02:~$ █

```

Task 5

```
student@6312vm-csr02:~/webserver$ curl http://172.17.0.2:8000
172.17.0.1 - - [06/May/2026 17:20:01] "GET / HTTP/1.1" 200 -
<!DOCTYPE HTML>
<html lang="en">
<head>
<meta charset="utf-8">
<style type="text/css">
:root {
color-scheme: light dark;
}
</style>
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href=".dockerenv">.dockerenv</a></li>
<li><a href=".rock/">.rock</a></li>
<li><a href="bin/">bin@</a></li>
<li><a href="boot/">boot</a></li>
<li><a href="dev/">dev</a></li>
<li><a href="etc/">etc</a></li>
<li><a href="home/">home</a></li>
<li><a href="lib/">lib@</a></li>
<li><a href="lib64/">lib64@</a></li>
<li><a href="media/">media</a></li>
<li><a href="mnt/">mnt</a></li>
<li><a href="opt/">opt</a></li>
<li><a href="proc/">proc</a></li>
<li><a href="root/">root</a></li>
<li><a href="run/">run</a></li>
<li><a href="sbin/">sbin@</a></li>
<li><a href="srv/">srv</a></li>
<li><a href="sys/">sys</a></li>
<li><a href="tmp/">tmp</a></li>
<li><a href="usr/">usr</a></li>
<li><a href="var/">var</a></li>
</ul>
<hr>
</body>
</html>
student@6312vm-csr02:~/webserver$ █
```


Task 6

```
student@6312vm-csr02:~$ git clone -b v1 https://github.com/inwk6312course/dockerapp.git
Cloning into 'dockerapp'...
remote: Enumerating objects: 48, done.
remote: Counting objects: 100% (48/48), done.
remote: Compressing objects: 100% (37/37), done.
remote: Total 48 (delta 15), reused 31 (delta 8), pack-reused 0 (from 0)
Unpacking objects: 100% (48/48), 8.98 KiB | 219.00 KiB/s, done.
student@6312vm-csr02:~$ cd dockerapp
student@6312vm-csr02:~/dockerapp$ ls
app Dockerfile LICENSE README.md
student@6312vm-csr02:~/dockerapp$ docker build -t dockerapp:v1 .
Sending build context to Docker daemon 131.1kB
Step 1/7 : FROM python:3.5
3.5: Pulling from library/python
57dflalflad8: Pull complete
71e126169501: Pull complete
1af28a55c3f3: Pull complete
03f1c9932170: Pull complete
65b3db15f518: Pull complete
850581be87f3: Pull complete
1e37775630ae: Pull complete
7e054ca5fcba: Pull complete
92a0fe226896: Pull complete
Digest: sha256:42a37d6b8c00b186bdfb2b620fa8023eb775b3eb3a768fd3c2e421964eee9665
Status: Downloaded newer image for python:3.5
---> 3687eb5ea744
Step 2/7 : RUN pip install Flask==2.3.0
---> Running in a94ef6398a03
DEPRECATION: Python 3.5 reached the end of its life on September 13th, 2020. Please upgrade your Python as Python 3.
5 is no longer maintained. pip 21.0 will drop support for Python 3.5 in January 2021. pip 21.0 will remove support f
or this functionality.
WARNING: You are using pip version 20.2.3; however, version 20.3.4 is available.
You should consider upgrading via the '/usr/local/bin/python -m pip install --upgrade pip' command.
Removing intermediate container a94ef6398a03
---> 4c142bb795a5
Step 3/7 : RUN useradd -ms /bin/bash admin
---> Running in 7c4f72a4cdc8
Removing intermediate container 7c4f72a4cdc8
---> adbb6d45c742
Step 4/7 : USER admin
---> Running in 48cc9f11f080
Removing intermediate container 48cc9f11f080
---> d02815819518
Step 5/7 : WORKDIR /app
---> Running in 40763e57683d
Removing intermediate container 40763e57683d
---> a98154ade6a8
Step 6/7 : COPY app /app
---> 2d8742fe5e6d
Step 7/7 : CMD ["python", "app.py"]
---> Running in 4668dfc572b7
Removing intermediate container 4668dfc572b7
---> 1326a3a3a278
Successfully built 1326a3a3a278
Successfully tagged dockerapp:v1
student@6312vm-csr02:~/dockerapp$ docker image ls
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
dockerapp            v1           1326a3a3a278     41 seconds ago   882MB
```

```
student@6312vm-csr02:~/dockerapp$ docker image ls
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
dockerapp            v1           1326a3a3a278     41 seconds ago   882MB
webserver            latest       9e221dfcccbe     6 minutes ago    181MB
<none>               <none>       31d61f0fde13     16 minutes ago   181MB
<none>               <none>       2b54951e0061     23 minutes ago   135MB
linux-lab_ubuntu     latest       ed1bbc491458     5 days ago       335MB
ubuntu               latest       30ba44506a6d     2 weeks ago      100MB
ubuntu               18.04       f9a80a55f492     2 years ago      63.2MB
python               3.5         3687eb5ea744     5 years ago      871MB
student@6312vm-csr02:~/dockerapp$ docker run -d -p 5000:5000 1326a3a3a278
483cde8f16852d208c2e3f3fc4fc2e37f24418e76036943722102b405939ea3a
student@6312vm-csr02:~/dockerapp$ curl localhost:5000
Hello, World!student@6312vm-csr02:~/dockerapp$
```

Task 7

```
student@6312vm-csr02:~/dockerapp$ docker container ls
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
376c4d6688a7   dockerapp:v2                        "python app.py"         2 minutes ago Up 2 minutes  0.0.0.0:5000->5000/tcp, ::
:5000->5000/tcp   inspiring_mcclintock
71c58f3b3a3e   redis:3.2.0                         "docker-entrypoint.s..." 4 minutes ago Up 4 minutes  6379/tcp
redis
540a22252d19   linux-lab_ubuntu                    "/bin/bash"             4 days ago    Up 4 days
my-ubuntu
student@6312vm-csr02:~/dockerapp$ docker exec -it 376c4d6688a7 bash
admin@376c4d6688a7:/app$ more /etc/hosts
127.0.0.1      localhost
::1           localhost ip6-localhost ip6-loopback
fe00::0       ip6-localnet
ff00::0       ip6-mcastprefix
ff02::1       ip6-allnodes
ff02::2       ip6-allrouters
172.17.0.2     redis 71c58f3b3a3e
172.17.0.3     376c4d6688a7
admin@376c4d6688a7:/app$ ping 172.17.0.3
PING 172.17.0.3 (172.17.0.3) 56(84) bytes of data.
64 bytes from 172.17.0.3: icmp_seq=1 ttl=64 time=0.064 ms
64 bytes from 172.17.0.3: icmp_seq=2 ttl=64 time=0.071 ms
64 bytes from 172.17.0.3: icmp_seq=3 ttl=64 time=0.079 ms
^C
--- 172.17.0.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 13ms
rtt min/avg/max/mdev = 0.064/0.071/0.079/0.009 ms
admin@376c4d6688a7:/app$
```

Task 8

```
redis_1        | 1:M 06 May 2026 17:44:24.354 * <search> Acquired RedisJSON_V6 API
redis_1        | 1:M 06 May 2026 17:44:24.355 * Server initialized
redis_1        | 1:M 06 May 2026 17:44:24.358 * Ready to accept connections tcp
redis_1        | 1:M 06 May 2026 17:44:24.358 # WARNING: Redis does not require authentication and is not protected by
network restrictions. Redis will accept connections from any IP address on any network interface.
dockerapp_1    | * Serving Flask app 'app'
dockerapp_1    | * Debug mode: off
dockerapp_1    | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSG
I server instead.
dockerapp_1    | * Running on all addresses (0.0.0.0)
dockerapp_1    | * Running on http://127.0.0.1:5000
dockerapp_1    | * Running on http://172.21.0.3:5000
dockerapp_1    | Press CTRL+C to quit
^CGracefully stopping... (press Ctrl+C again to force)
Stopping dockerapp_dockerapp_1 ... done
Stopping dockerapp_redis_1 ... done
student@6312vm-csr02:~/dockerapp$ docker-compose up -d
Starting dockerapp_redis_1 ... done
Starting dockerapp_dockerapp_1 ... done
student@6312vm-csr02:~/dockerapp$ curl localhost:5000
<html>
<head>
<title>key value lookup service</title>
</head>
<body>
<form method="POST">
<input type="text" name="key" value=1>
<input type="text" name="cache_value" value=one>
<input type="submit" name="submit" value="load">
<input type="submit" name="submit" value="save">
</form>
</body>
</html>student@6312vm-csr02:~/dockerapp$
```

```

redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> Redis version found by RedisTimeSeries : 8.6.3 - oss
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> Registering configuration options: [
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> { ts-compaction-policy : }
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> { ts-num-threads : 3 }
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> { ts-retention-policy : 0 }
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> { ts-duplicate-policy : block }
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> { ts-chunk-size-bytes : 4096 }
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> { ts-encoding : compressed }
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> { ts-ignore-max-time-diff: 0 }
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> { ts-ignore-max-val-diff : 0.000000 }
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> ]
redis_1 | 1:M 06 May 2026 17:44:57.410 * <timeseries> Detected redis oss
redis_1 | 1:M 06 May 2026 17:44:57.411 * <timeseries> Subscribe to ASM events
redis_1 | 1:M 06 May 2026 17:44:57.411 * <timeseries> Enabled diskless replication
redis_1 | 1:M 06 May 2026 17:44:57.411 * Module 'timeseries' loaded from /usr/local/lib/redis/modules//redistim
eseries.so
redis_1 | 1:M 06 May 2026 17:44:57.411 * <ReJSON> Created new data type 'ReJSON-RL'
redis_1 | 1:M 06 May 2026 17:44:57.412 * <ReJSON> version: 80600 git sha: unknown branch: unknown
redis_1 | 1:M 06 May 2026 17:44:57.412 * <ReJSON> Exported RedisJSON_V1 API
redis_1 | 1:M 06 May 2026 17:44:57.412 * <ReJSON> Exported RedisJSON_V2 API
redis_1 | 1:M 06 May 2026 17:44:57.412 * <ReJSON> Exported RedisJSON_V3 API
redis_1 | 1:M 06 May 2026 17:44:57.412 * <ReJSON> Exported RedisJSON_V4 API
redis_1 | 1:M 06 May 2026 17:44:57.412 * <ReJSON> Exported RedisJSON_V5 API
redis_1 | 1:M 06 May 2026 17:44:57.412 * <ReJSON> Exported RedisJSON_V6 API
redis_1 | 1:M 06 May 2026 17:44:57.412 * <ReJSON> Enabled diskless replication
redis_1 | 1:M 06 May 2026 17:44:57.412 * <ReJSON> Initialized shared string cache, thread safe: true.
redis_1 | 1:M 06 May 2026 17:44:57.412 * Module 'ReJSON' loaded from /usr/local/lib/redis/modules//rejson.so
redis_1 | 1:M 06 May 2026 17:44:57.412 * <search> Acquired RedisJSON_V6 API
redis_1 | 1:M 06 May 2026 17:44:57.412 * Server initialized
redis_1 | 1:M 06 May 2026 17:44:57.413 * <search> Changing workers threadpool size from 0 to 4
redis_1 | 1:M 06 May 2026 17:44:57.413 * <search> Enabled workers threadpool of size 4
redis_1 | 1:M 06 May 2026 17:44:57.413 * <search> Loading event started
redis_1 | 1:M 06 May 2026 17:44:57.414 * Loading RDB produced by version 8.6.3
redis_1 | 1:M 06 May 2026 17:44:57.414 * RDB age 15 seconds
redis_1 | 1:M 06 May 2026 17:44:57.414 * RDB memory usage when created 1.31 Mb
redis_1 | 1:M 06 May 2026 17:44:57.414 * Done loading RDB, keys loaded: 1, keys expired: 0.
redis_1 | 1:M 06 May 2026 17:44:57.414 * <search> Changing workers threadpool size from 4 to 0
redis_1 | 1:M 06 May 2026 17:44:57.414 * <search> Scheduling config_reduce_threads_job to remove all 4 threads
when empty
redis_1 | 1:M 06 May 2026 17:44:57.414 * <search> Disabled workers threadpool of size 4
redis_1 | 1:M 06 May 2026 17:44:57.414 * <search> Loading event ended successfully
redis_1 | 1:M 06 May 2026 17:44:57.414 * DB loaded from disk: 0.002 seconds
redis_1 | 1:M 06 May 2026 17:44:57.414 * Ready to accept connections tcp
redis_1 | 1:M 06 May 2026 17:44:57.414 # WARNING: Redis does not require authentication and is not protected by
network restrictions. Redis will accept connections from any IP address on any network interface.
student@6312vm-csr02:~/dockerapp$ docker-compose ps

```

Name	Command	State	Ports
dockerapp dockerapp_1	python app.py	Up	0.0.0.0:5000->5000/tcp, :::5000->5000/tcp
dockerapp redis_1	docker-entrypoint.sh redis ...	Up	6379/tcp

Task 9

```
student@6312vm-csr02:~$  
student@6312vm-csr02:~$ docker network ls  
NETWORK ID          NAME                DRIVER              SCOPE  
2852201907dc        bridge              bridge              local  
91940715fee8         host                host                local  
e53c32371d22         linux-lab_default   bridge              local  
b6535e8a1727         none                null                local  
student@6312vm-csr02:~$
```

```
student@6312vm-csr02:~$ docker run -d --net none busybox sleep 1000  
Unable to find image 'busybox:latest' locally  
latest: Pulling from library/busybox  
481282afbc43: Pull complete  
Digest: sha256:1487d0af5f52b4ba31c7e465126ee2123fe3f2305d638e7827681e7cf6c83d5e  
Status: Downloaded newer image for busybox:latest  
3610fbe8ac35b72844b50ccaa557fae64a0e4d5aeaff9b34df98fdf6c459d0bd  
student@6312vm-csr02:~$ docker ps -a  
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS              NAMES  
3610fbe8ac35        busybox            "sleep 1000"        7 seconds ago       Up 5 seconds                    quirky_ro  
binson  
4753f48144d0        31d61f0fde13      "/helloinwk.py"     43 minutes ago      Exited (0) 43 minutes ago        cranky_br  
attain  
4eb6fba9c250        2b54951e0061       "/bin/bash"         50 minutes ago      Exited (0) 50 minutes ago        admiring_  
nightingale  
540a22252d19        linux-lab_ubuntu   "/bin/bash"         5 days ago          Up 5 days                    my-ubuntu  
student@6312vm-csr02:~$ docker exec -it 3610fbe8ac35 /bin/ash  
/ # ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8): 56 data bytes  
ping: sendto: Network is unreachable  
/ # ifconfig  
lo                Link encap:Local Loopback  
inet addr:127.0.0.1 Mask:255.0.0.0  
UP LOOPBACK RUNNING MTU:65536 Metric:1  
RX packets:0 errors:0 dropped:0 overruns:0 frame:0  
TX packets:0 errors:0 dropped:0 overruns:0 carrier:0  
collisions:0 txqueuelen:1000  
RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)  
  
/ #
```

```

student@6312vm-csr02:~$ docker container ls
CONTAINER ID   IMAGE          COMMAND          CREATED          STATUS          PORTS          NAMES
3610fbe8ac35   busybox        "sleep 1000"     About a minute ago   Up About a minute
540a22252d19   linux-lab_ubuntu "/bin/bash"      5 days ago       Up 5 days       my-ubuntu
student@6312vm-csr02:~$ docker container stop 3610fbe8ac35
3610fbe8ac35
student@6312vm-csr02:~$ docker container rm 3610fbe8ac35
3610fbe8ac35
student@6312vm-csr02:~$ docker container ls
CONTAINER ID   IMAGE          COMMAND          CREATED          STATUS          PORTS          NAMES
540a22252d19   linux-lab_ubuntu "/bin/bash"      5 days ago       Up 5 days       my-ubuntu
student@6312vm-csr02:~$ docker run -d --name container1 busybox sleep 1000
05cf761f5243690ef7408d3cdc71733983295b61a78d5c7dadedcaac7166d796
student@6312vm-csr02:~$ docker container ls
CONTAINER ID   IMAGE          COMMAND          CREATED          STATUS          PORTS          NAMES
05cf761f5243   busybox        "sleep 1000"     11 seconds ago   Up 10 seconds   container1
540a22252d19   linux-lab_ubuntu "/bin/bash"      5 days ago       Up 5 days       my-ubuntu
student@6312vm-csr02:~$ docker exec -it container1 ifconfig
eth0      Link encap:Ethernet HWaddr 02:42:AC:11:00:02
          inet addr:172.17.0.2 Bcast:172.17.255.255 Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
          RX packets:44 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:7740 (7.5 KiB) TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1 Mask:255.0.0.0
          UP LOOPBACK RUNNING MTU:65536 Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

student@6312vm-csr02:~$

```

```

student@6312vm-csr02:~$ docker run -d --name container2 busybox sleep 1000
0caa097a9d2967827aab023bbc5ae2e1e7f51dfb81cfa086c9692d972b92f269
student@6312vm-csr02:~$ docker exec -it container2 ifconfig
eth0      Link encap:Ethernet HWaddr 02:42:AC:11:00:03
          inet addr:172.17.0.3 Bcast:172.17.255.255 Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
          RX packets:18 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:2415 (2.3 KiB) TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1 Mask:255.0.0.0
          UP LOOPBACK RUNNING MTU:65536 Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B) TX bytes:0 (0.0 B)

student@6312vm-csr02:~$

```

```
student@6312vm-csr02:~$ docker exec -it container1 ping 172.17.0.3
PING 172.17.0.3 (172.17.0.3): 56 data bytes
64 bytes from 172.17.0.3: seq=0 ttl=64 time=0.677 ms
64 bytes from 172.17.0.3: seq=1 ttl=64 time=0.154 ms
64 bytes from 172.17.0.3: seq=2 ttl=64 time=0.165 ms
64 bytes from 172.17.0.3: seq=3 ttl=64 time=0.165 ms
64 bytes from 172.17.0.3: seq=4 ttl=64 time=0.156 ms
64 bytes from 172.17.0.3: seq=5 ttl=64 time=0.160 ms
64 bytes from 172.17.0.3: seq=6 ttl=64 time=0.150 ms
^C
--- 172.17.0.3 ping statistics ---
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/max = 0.150/0.232/0.677 ms
student@6312vm-csr02:~$ docker exec -it container1 ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: seq=0 ttl=116 time=15.465 ms
64 bytes from 8.8.8.8: seq=1 ttl=116 time=15.431 ms
64 bytes from 8.8.8.8: seq=2 ttl=116 time=15.278 ms
64 bytes from 8.8.8.8: seq=3 ttl=116 time=15.389 ms
64 bytes from 8.8.8.8: seq=4 ttl=116 time=15.318 ms
64 bytes from 8.8.8.8: seq=5 ttl=116 time=15.313 ms
64 bytes from 8.8.8.8: seq=6 ttl=116 time=15.354 ms
^C
--- 8.8.8.8 ping statistics ---
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/max = 15.278/15.364/15.465 ms
student@6312vm-csr02:~$
```

```
student@6312vm-csr02:~$ docker exec -it container3 ping 172.17.0.3
PING 172.17.0.3 (172.17.0.3): 56 data bytes
^C
--- 172.17.0.3 ping statistics ---
8 packets transmitted, 0 packets received, 100% packet loss
student@6312vm-csr02:~$
```


Task 10

```
student@6312vm-csr02:~$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
2852201907dc        bridge              bridge               local
91940715fee8         host                host                 local
e53c32371d22         linux-lab_default   bridge               local
85d2035ad895         my_bridge_net       bridge               local
b6535e8a1727         none                null                 local
student@6312vm-csr02:~$ docker network inspect my_bridge_net
[
  {
    "Name": "my_bridge_net",
    "Id": "85d2035ad895ccea36098f1e47ac7fca3bc3b3913da137f5e459611555deae9c",
    "Created": "2026-05-06T15:00:38.371043759-03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.22.0.0/16",
          "Gateway": "172.22.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
student@6312vm-csr02:~$ docker run -d --name container3 --net my_bridge_net busybox sleep 1000
7f6c360d9768c0a2b3df0b63c64bb9b9c2038a48a6838ela450f25fc04f96c81
student@6312vm-csr02:~$ docker exec -it container3 ifconfig
eth0      Link encap:Ethernet  HWaddr 02:42:AC:16:00:02
          inet addr:172.22.0.2  Bcast:172.22.255.255  Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:45 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:6430 (6.2 KiB)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

student@6312vm-csr02:~$
```

Task 11

```
student@6312vm-csr02:~$ docker network connect bridge container3
student@6312vm-csr02:~$ docker exec -it container3 ifconfig
eth0      Link encap:Ethernet  HWaddr 02:42:AC:16:00:02
          inet addr:172.22.0.2  Bcast:172.22.255.255  Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:227 errors:0 dropped:0 overruns:0 frame:0
          TX packets:9 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:45960 (44.8 KiB)  TX bytes:826 (826.0 B)

eth1      Link encap:Ethernet  HWaddr 02:42:AC:11:00:04
          inet addr:172.17.0.4  Bcast:172.17.255.255  Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:18 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:2415 (2.3 KiB)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

student@6312vm-csr02:~$ docker exec -it container3 ping 172.17.0.2
PING 172.17.0.2 (172.17.0.2): 56 data bytes
64 bytes from 172.17.0.2: seq=0 ttl=64 time=0.232 ms
64 bytes from 172.17.0.2: seq=1 ttl=64 time=0.169 ms
64 bytes from 172.17.0.2: seq=2 ttl=64 time=0.185 ms
64 bytes from 172.17.0.2: seq=3 ttl=64 time=0.163 ms
64 bytes from 172.17.0.2: seq=4 ttl=64 time=0.160 ms
64 bytes from 172.17.0.2: seq=5 ttl=64 time=0.169 ms
64 bytes from 172.17.0.2: seq=6 ttl=64 time=0.168 ms
^C
--- 172.17.0.2 ping statistics ---
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/max = 0.160/0.178/0.232 ms
student@6312vm-csr02:~$
```

Task 12

```
student@6312vm-csr02:~$ docker run -d --name container4 --net host busybox sleep 1000
dadbb4d973bbaca95537c241a56bc709b99783fde9fa51ac6a3bcc3e37349de8
student@6312vm-csr02:~$ docker exec -it container4 ifconfig
br-85d2035ad895 Link encap:Ethernet HWaddr 02:42:9A:7E:FS:3A
    inet addr:172.22.0.1 Bcast:172.22.255.255 Mask:255.255.0.0
    inet6 addr: fe80::42:9aff:fe7e:f53a/64 Scope:Link
    UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
    RX packets:9 errors:0 dropped:0 overruns:0 frame:0
    TX packets:328 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:0
    RX bytes:700 (700.0 B) TX bytes:73157 (71.4 KiB)

br-e53c32371d22 Link encap:Ethernet HWaddr 02:42:6F:68:44:1C
    inet addr:172.20.0.1 Bcast:172.20.255.255 Mask:255.255.0.0
    inet6 addr: fe80::42:6fff:fe68:441c/64 Scope:Link
    UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
    RX packets:57036 errors:0 dropped:0 overruns:0 frame:0
    TX packets:379230 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:0
    RX bytes:8065715 (7.6 MiB) TX bytes:229157627 (218.5 MiB)

docker0 Link encap:Ethernet HWaddr 02:42:99:7C:5A:4F
    inet addr:172.17.0.1 Bcast:172.17.255.255 Mask:255.255.0.0
    inet6 addr: fe80::42:99ff:fe7c:5a4f/64 Scope:Link
    UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
    RX packets:30626 errors:0 dropped:0 overruns:0 frame:0
    TX packets:49558 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:0
    RX bytes:1682818 (1.6 MiB) TX bytes:182613013 (174.1 MiB)

ens160 Link encap:Ethernet HWaddr 00:50:56:A8:09:E4
    inet addr:10.0.134.102 Bcast:10.0.134.255 Mask:255.255.255.0
    inet6 addr: fe80::24af:a0d7:d080:cdb2/64 Scope:Link
    inet6 addr: fe80::c72b:f823:5101:157c/64 Scope:Link
    inet6 addr: fe80::32eb:39e5:c0b4:5cc6/64 Scope:Link
    UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
    RX packets:3823958 errors:0 dropped:455 overruns:0 frame:0
    TX packets:787562 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:1000
    RX bytes:3574985488 (3.3 GiB) TX bytes:70015421 (66.7 MiB)

ens192 Link encap:Ethernet HWaddr 00:50:56:A8:AF:60
    inet addr:192.168.1.20 Bcast:192.168.1.255 Mask:255.255.255.0
    inet6 addr: fe80::7e9f:1a61:218:b5df/64 Scope:Link
    UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
    RX packets:79 errors:0 dropped:0 overruns:0 frame:0
    TX packets:15998 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:1000
    RX bytes:8937 (8.7 KiB) TX bytes:3764584 (3.5 MiB)

lo Link encap:Local Loopback
    inet addr:127.0.0.1 Mask:255.0.0.0
    inet6 addr: ::1/128 Scope:Host
    UP LOOPBACK RUNNING MTU:65536 Metric:1
    RX packets:24400 errors:0 dropped:0 overruns:0 frame:0
    TX packets:24400 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:1000
    RX bytes:4353600 (4.1 MiB) TX bytes:4353600 (4.1 MiB)
```

```
lo          Link encap:Local Loopback
            inet addr:127.0.0.1  Mask:255.0.0.0
            inet6 addr: ::1/128 Scope:Host
            UP LOOPBACK RUNNING  MTU:65536  Metric:1
            RX packets:24400 errors:0 dropped:0 overruns:0 frame:0
            TX packets:24400 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:1000
            RX bytes:4353600 (4.1 MiB)  TX bytes:4353600 (4.1 MiB)

vethcced607 Link encap:Ethernet  HWaddr 46:DD:EA:39:FA:A7
            inet6 addr: fe80::44dd:eaff:fe39:faa7/64 Scope:Link
            UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
            RX packets:9 errors:0 dropped:0 overruns:0 frame:0
            TX packets:504 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:0
            RX bytes:826 (826.0 B)  TX bytes:108484 (105.9 KiB)

vetheaad551 Link encap:Ethernet  HWaddr 2A:30:2E:B5:14:D2
            inet6 addr: fe80::2830:2eff:feb5:14d2/64 Scope:Link
            UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
            RX packets:57036 errors:0 dropped:0 overruns:0 frame:0
            TX packets:530513 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:0
            RX bytes:8864219 (8.4 MiB)  TX bytes:261973546 (249.8 MiB)

vethef34aba Link encap:Ethernet  HWaddr 0A:BC:47:ED:F3:B4
            inet6 addr: fe80::8bc:47ff:feed:f3b4/64 Scope:Link
            UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
            RX packets:27 errors:0 dropped:0 overruns:0 frame:0
            TX packets:837 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:0
            RX bytes:2310 (2.2 KiB)  TX bytes:183051 (178.7 KiB)

vethf3cd049 Link encap:Ethernet  HWaddr EE:69:F4:F5:59:D5
            inet6 addr: fe80::ec69:f4ff:fef5:59d5/64 Scope:Link
            UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
            RX packets:9 errors:0 dropped:0 overruns:0 frame:0
            TX packets:735 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:0
            RX bytes:770 (770.0 B)  TX bytes:162211 (158.4 KiB)

student@6312vm-csr02:~$ █
```

Task 13

```
redis_1 | 1:M 06 May 2026 18:11:15.132 * Ready to accept connections tcp
redis_1 | 1:M 06 May 2026 18:11:15.132 # WARNING: Redis does not require authentication and is not protected by
network restrictions. Redis will accept connections from any IP address on any network interface.
dockerapp_1 | * Serving Flask app 'app'
dockerapp_1 | * Debug mode: off
dockerapp_1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSG
I server instead.
dockerapp_1 | * Running on all addresses (0.0.0.0)
dockerapp_1 | * Running on http://127.0.0.1:5000
dockerapp_1 | * Running on http://172.23.0.3:5000
dockerapp_1 | Press CTRL+C to quit
^CGracefully stopping... (press Ctrl+C again to force)
Stopping dockerapp_dockerapp_1 ... done
Stopping dockerapp_redis_1 ... done
student@6312vm-csr02:~/dockerapp$ docker-compose up -d
Starting dockerapp_redis_1 ... done
Starting dockerapp_dockerapp_1 ... done
student@6312vm-csr02:~/dockerapp$ docker-compose ps
      Name                                Command                                State                  Ports
-----
dockerapp_dockerapp_1  python app.py                        Up                    0.0.0.0:5000->5000/tcp, :::5000->5000/tcp
dockerapp_redis_1     docker-entrypoint.sh redis ...      Up                    6379/tcp
student@6312vm-csr02:~/dockerapp$
```

```
student@6312vm-csr02:~/dockerapp$ docker-compose ps
      Name                                Command                                State                  Ports
-----
dockerapp_dockerapp_1  python app.py                        Up                    0.0.0.0:5000->5000/tcp, :::5000->5000/tcp
dockerapp_redis_1     docker-entrypoint.sh redis ...      Up                    6379/tcp
student@6312vm-csr02:~/dockerapp$ curl localhost:5000
<html>
  <head>
    <title>key value lookup service</title>
  </head>
  <body>
    <form method="POST">
      <input type="text" name="key" value=l>
      <input type="text" name="cache_value" value=one>
      <input type="submit" name="submit" value="load">
      <input type="submit" name="submit" value="save">
    </form>
  </body>
</html>student@6312vm-csr02:~/dockerapp$
```